Superstore Analysis
DIY Project 3

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Problem Statement of Superstore Analysis

Retail businesses generate vast amounts of transactional data across products, regions, and customer segments, yet many struggle to translate this raw data into actionable business insight. A superstore chain operating across multiple U.S. regions sells products across three broad categories — Furniture, Office Supplies, and Technology — to Consumer, Corporate, and Home Office customer segments, using varying shipping modes (Standard Class, Second Class, First Class, Same Day).

While the company records detailed order-level data including sales, quantity, discount, and profit, it currently lacks a clear understanding of which factors are driving profitability and which are eroding it. Preliminary patterns suggest that certain sub-categories (like Tables and Bookcases) and high-discount transactions are associated with significant losses, even when sales figures appear strong. Without a systematic analysis, the company cannot answer key questions such as:

Which products and regions are most profitable, and which are underperforming?

How does discounting affect profit margins, and at what point does it become unsustainable?

Which customer segments and shipping modes contribute most to revenue and profit?

Are there regional or state-level disparities in performance that need strategic attention?

This lack of visibility limits the company's ability to make informed decisions on pricing, discounting policy, inventory focus, and regional resource allocation.

Project Description of this Project

This project involves conducting a comprehensive Exploratory Data Analysis (EDA) on the Superstore dataset to uncover sales and profitability trends across products, regions, customer segments, and shipping modes.
This Dataset Includes

Ship Mode : The method used to deliver the order, such as Standard Class, Second Class, First Class, or Same Day.

Segment :The type of customer, such as Consumer, Corporate, or Home Office.

Country: The country where the customer or order is located.

City: The city where the order was placed or delivered.

State: The state where the customer or order is located.

Postal Code: The postal or ZIP code of the customer's location.

Region: The geographical region, such as Central, East, South, or West.

Category: The main product category, such as Furniture, Office Supplies, or Technology.

Sub-Category: A more specific product group within a category, such as Chairs, Phones, Binders, or Storage.

Sales: The total sales amount generated from the order.

Quantity :The number of products or items sold in the order

Discount: The discount percentage or rate given on the product or order.

Profit: The profit or loss earned from the order after considering sales and related costs. A positive value indicates profit, while a negative value indicates loss

Project Objectives

1) Which region has the highest total sales?

2) Which region generates the highest profit?

3) Which category has the highest total sales

4) Which category has the highest Profit ?

5) Which sub-category generates the highest Sales?

6) Which sub-category generates the highest Profit?

7) Which sub-category has the lowest or negative profit?

Which customer segment generates the most sales?

8) Which Ship mode has the highest sales

9) How does discount affect profit?

10) What is the relationship between Sales, Quantity, Discount, and Profit?

What is Data Wrangling ?

Data wrangling, a crucial phase in the data analysis process, involves preparing and cleaning raw data to make it suitable for analysis. This involves various tasks such as handling missing values, removing duplicates, and transforming data into a consistent format. Effective data wrangling ensures that the data is reliable and ready for exploration, enabling meaningful insights to be derived from it.

What is the Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA)is a vital phase in data analysis involving the examination and visualization of data to uncover patterns ,trends, and insights.in the context of an Airbnb data analysis project, EDA allows researchers to gain a Comprehensive understanding of the dataset's characteristics, distribution and relationships between variables. Through techniques such as summary stastics, data visualization ,and correlation analysis, EDA sets the stage for further investigation and hypothesis testing, uiding subsequent analytical decisions.

End Users of this Project

Superstore Analysis Project is useful for the following end users:

Business Managers - To get awareness about sales, profit, discounts and to understand business performance.

Sales Managers - To find top performing states, cities, region, segment, category, and sub-category.

Marketing Team - To study about different customer segments and to find the affect of discount on profit.

Finance Team - To track down the profit, sales and to find out the segments where discount results in loss to profit.

Business Analyst - To analyze trends and generate recommendations based on the data analysis.

Store/Operation Managers - To understand which products and in which states and regions are performing better.

Company Management - To use the analysis to build the business strategy and to improve overall company profitability.

Technologies Used In this Project

Hardware Requirements

The hardware requirements for performing the Superstore Analysis are:

RAM – 8.00 GB

Operating System – Windows 11

Processor – Intel(R) Core(TM) i3-1115g4

Processor speed – 3.00 GHz

Software Requirements

The software requirements for performing the Superstore Analysis are:

Programming Language – Python

Development Environment / IDE – Jupyter Notebook

Data Analysis Libraries – Pandas, NumPy

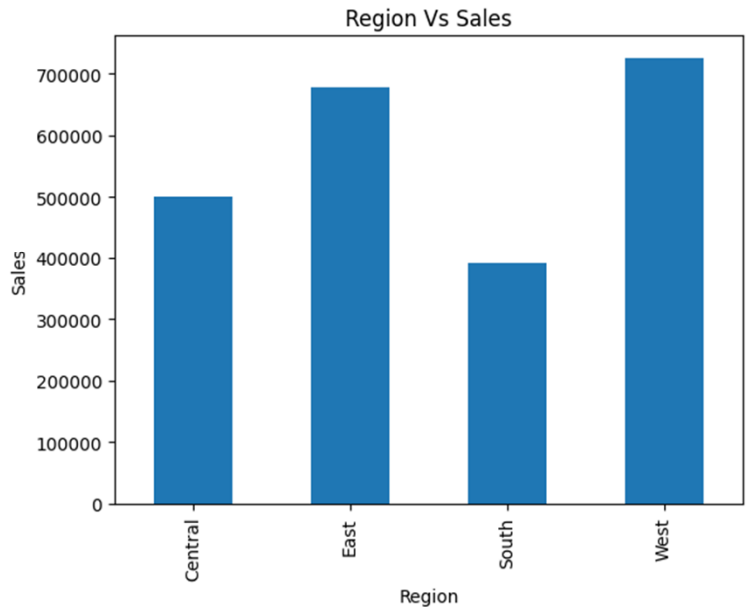
Data Visualization Libraries – Matplotlib, Seaborn

Dataset Format – CSV

Results of the Project

1) Which region has the highest total sales?

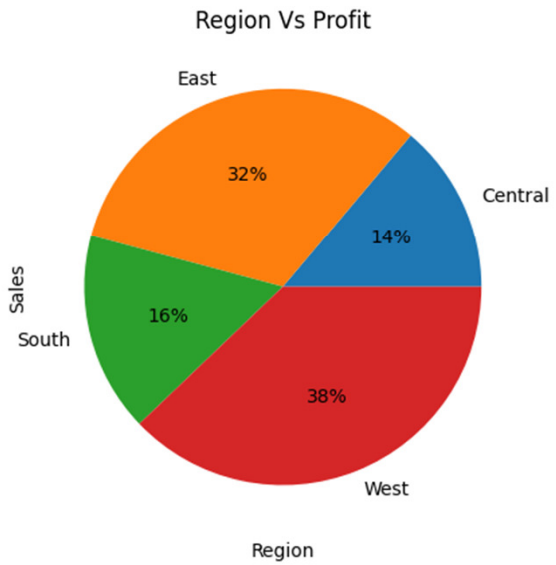
```
[24]: df.groupby("Region")["Sales"].sum().plot.bar()
      plt.title("Region Vs Sales")
      plt.xlabel("Region")
      plt.ylabel("Sales")
      plt.show()
```



West Region has the Highest Sales

2) Which region generates the highest profit?

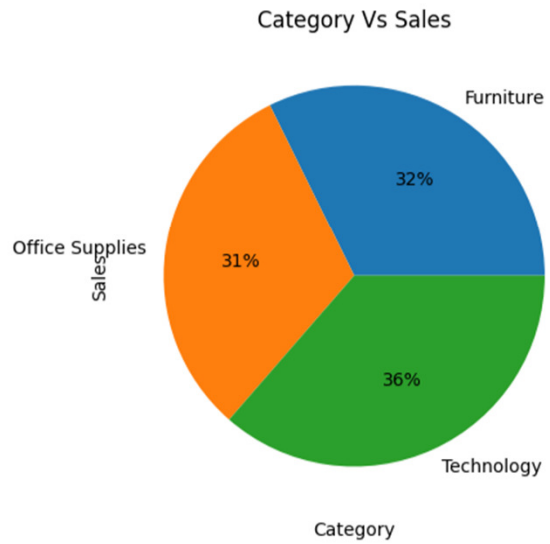
```
[55]: df.groupby("Region")["Profit"].sum().plot.pie(autopct="%1.0f%%")
      plt.title("Region Vs Profit")
      plt.xlabel("Region")
      plt.ylabel("Sales")
      plt.show()
```



West Region has the Highest Profit

3) Which category has the highest total sales ?

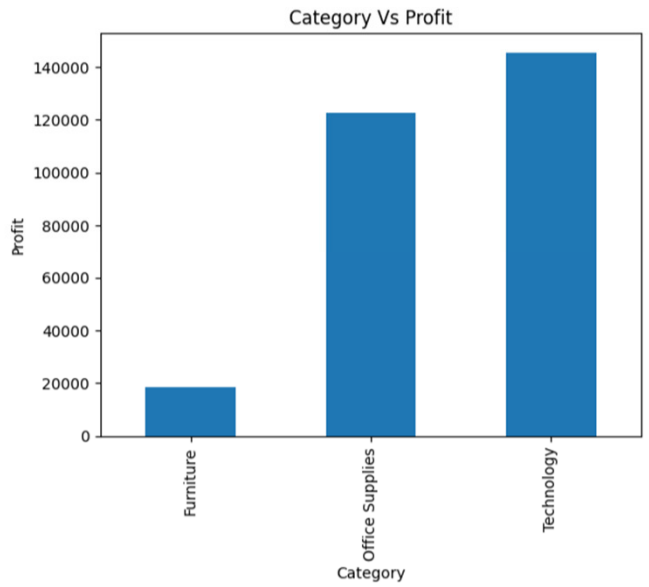
```
[43]: df.groupby("Category")["Sales"].sum().plot.pie(autopct="%1.0f%%")
      plt.title("Category Vs Sales")
      plt.xlabel("Category")
      plt.ylabel("Sales")
      plt.show()
```



Technology has the Highest Sales

4) Which category has the highest Profit ?

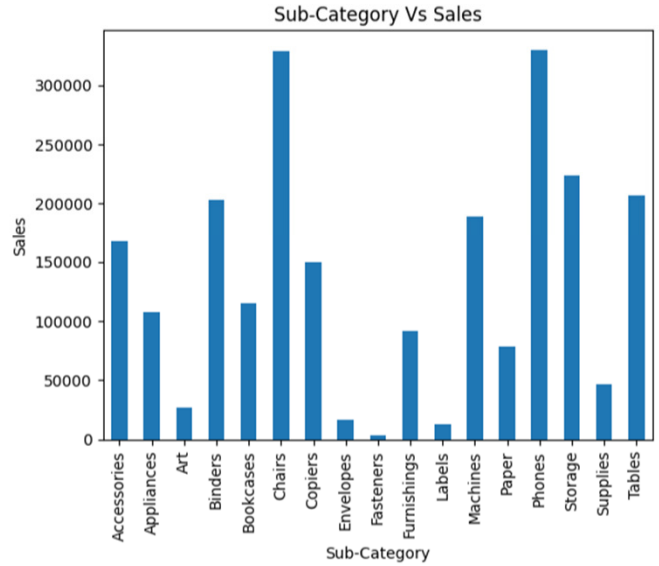
[57]:
```
df.groupby("Category")["Profit"].sum().plot.bar()
plt.title("Category Vs Profit")
plt.xlabel("Category")
plt.ylabel("Profit")
plt.show()
```



Technology has the Highest Profit

5) Which sub-category generates the highest Sales?

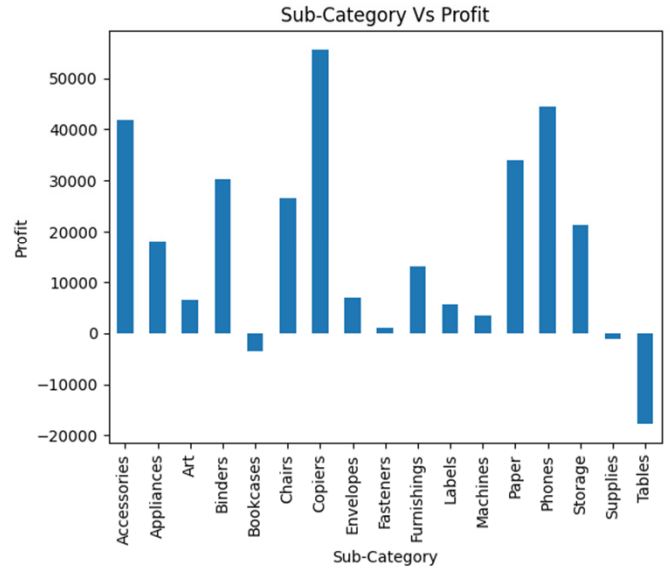
```
[45]: df.groupby("Sub-Category")["Sales"].sum().plot.bar()
      plt.title("Sub-Category Vs Sales")
      plt.xlabel("Sub-Category")
      plt.ylabel("Sales")
      plt.show()
```



Chair has the highest Sales

6) Which sub-category generates the highest Profit?

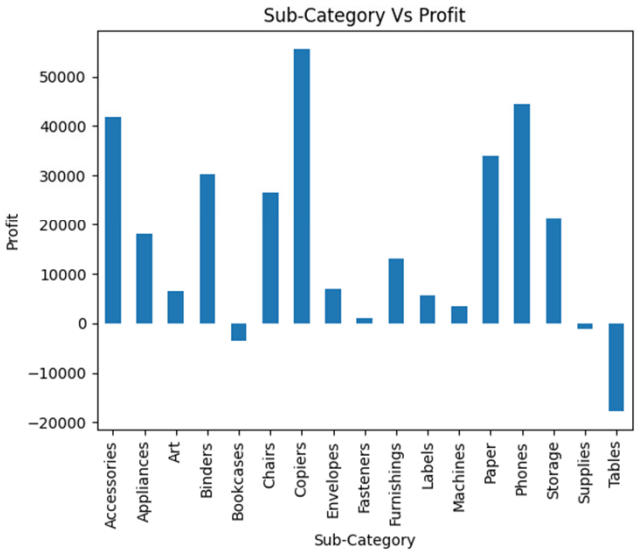
```
[88]: df.groupby("Sub-Category")["Profit"].sum().plot.bar()
      plt.title("Sub-Category Vs Profit")
      plt.xlabel("Sub-Category")
      plt.ylabel("Profit")
      plt.show()
```



Copiers has the Highest Profit

7) Which sub-category has the lowest or negative profit?

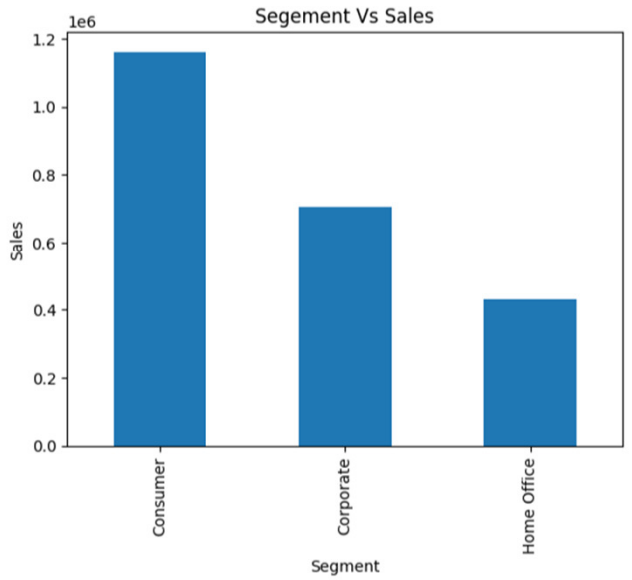
```
[88]: df.groupby("Sub-Category")["Profit"].sum().plot.bar()
      plt.title("Sub-Category Vs Profit")
      plt.xlabel("Sub-Category")
      plt.ylabel("Profit")
      plt.show()
```



Tables has the Negative Profit

8) Which customer segment generates the most sales?

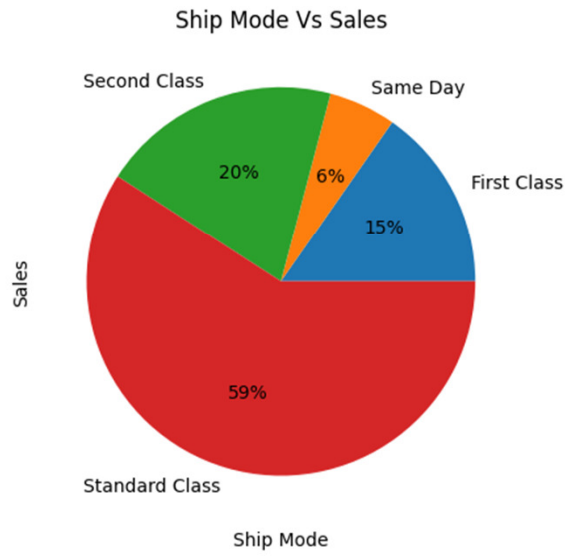
```
[66]: df.groupby("Segment")["Sales"].sum().plot.bar()
      plt.title("Segement Vs Sales")
      plt.xlabel("Segment")
      plt.ylabel("Sales")
      plt.show()
```



Consumer Segment generates the most sales

8) Which Ship mode has the highest sales ?

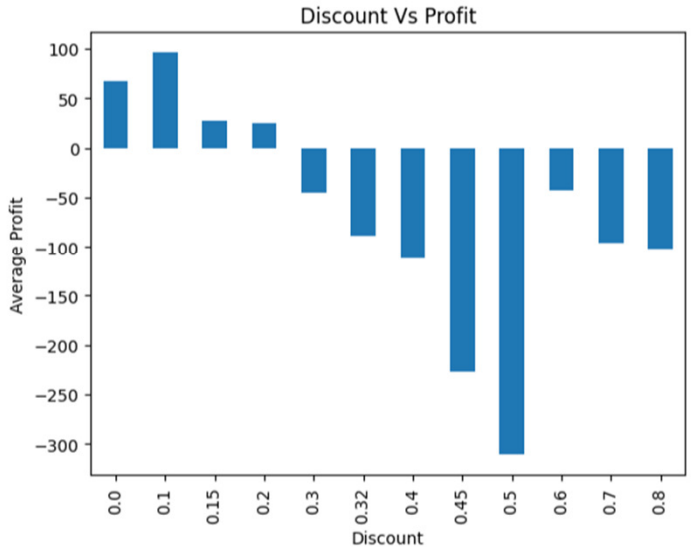
[67]:
```
df.groupby("Ship Mode")["Sales"].sum().plot.pie(autopct="%1.0f%%")
plt.title("Ship Mode Vs Sales")
plt.xlabel("Ship Mode")
plt.ylabel("Sales")
plt.show()
```



Standard Class ship mode has the highest sales

9) How does discount affect profit?

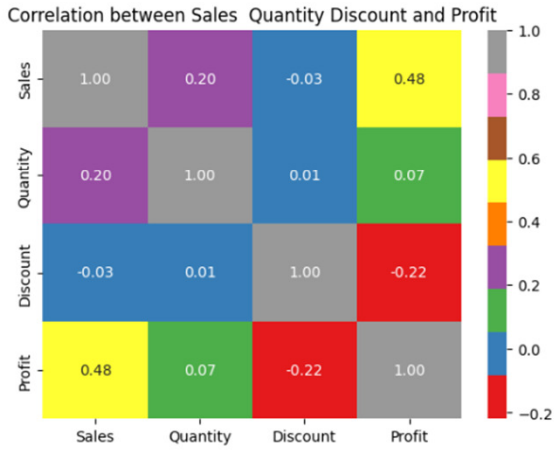
[70]:
```
df.groupby("Discount")["Profit"].mean().plot.bar()
plt.title("Discount Vs Profit")
plt.xlabel("Discount")
plt.ylabel("Average Profit")
plt.show()
```



The research demonstrates the impact that discount rates had on the average profit. It shows that discounts will lead to decrease in the profit, and at times could lead to negative profit thus it is not worthwhile for the product to have discounts if the profit margin of the product is low. This will benefit the organization by avoiding large discounts to products with low profits margin.

10) What is the relationship between Sales, Quantity, Discount, and Profit?

[94]:
```
import matplotlib.pyplot as plt
import seaborn as sns
correlation = df[["Sales", "Quantity", "Discount", "Profit"]].corr()
correlation = df.corr(numeric_only =True)
sns.heatmap (
    correlation,
    annot = True,
    cmap ='Set1',
    fmt=".2f"
)
plt.title("Correlation between Sales  Quantity Discount and Profit")
plt.show()
```



Sales and Profit have the strongest relationship (0.48) among these variables, meaning higher sales are generally associated with higher profit.

Discount and Profit have a negative correlation (-0.22), suggesting that higher discounts are generally associated with lower profit.

Quantity and Profit have a very weak correlation (0.07), meaning the number of items sold has very little linear relationship with profit in this dataset.

Future Scope

Predictive Analytics Integration: Go beyond a purely descriptive EDA. Build predictive models (regression, time series forecasting etc.) to predict future sales, demand and profit trends at a category, regional or seasonal level.

Customer Segmentation & Behaviour Analysis: Utilize clustering techniques (RFM analysis, K-Means etc.) on the customer data to group them by purchasing behaviour, allowing for a more refined and targeted marketing strategy.

Discount Optimization Model: Create a data-driven discount optimization or pricing model. This model could predict the optimal discount for each sub-category/region to maximise profit while staying competitive.

Inventory & Demand Planning :Integrate the analysis with current inventory data to find areas with overstocked/understocked products which should lead to better demand forecasting and improved supply chain management.

Real-Time Dashboard Automation :Transform this static analysis into a live automated dashboard (e.g. Power BI/Tableau pointing to a live database) that updates in real time as new sales come in.

Geospatial/Market Expansion Analysis: Use the location data to map sales and identify potential areas of business expansion. This could be at a state, city or even neighbourhood level.

Integration with External Data: Integrate this dataset with external data sources such as seasonality, economic indicators, competitor pricing etc. In order to capture additional insights into the drivers of sales and profit.

Customer Feedback & Sentiment Analysis: If customer review/feedback data is available, it can be analysed for sentiment to see if and how it is correlating with repeat business or profitability.

Conclusion

Overall, the sales throughout all the product categories as well as different regions look normal or strong. The profit however is highly distorted because too much discount was given throughout the company, especially in Tables and Bookcases sub-category where most of the sale recorded in those sub-categories given very large profit which means 50% and above discount is usually results in loss despite very strong sale. The variable performance among different region, customer type and the ship mode also suggest that business model has to take regional difference into account and a customized strategic decision, based on the data is the best approach to achieve higher profit. Based on the findings, the data used in this project has provided good evidence for business on deciding their strategies, by setting more reasonable discount strategy, emphasizing the right product at the right region with efficient strategy, improving customer engagement and analyzing those failing segments are all effective ways for the business to improve their overall profit, for retail, having insight data to generate strategies is a valuable factor to succeed.

Thank You

GitHub Link

https://github.com/krishnamohanb21-svg/Superstore-Analysis-DIY-Project-3-.git